

Jose Sanchez Gonzalez

Machine Learning Engineer / RAG and LLM Systems

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Summary

M.S. Artificial Intelligence building retrieval APIs, agent workflows, and inference services. Connects measured ranking quality to PostgreSQL/pgvector and uses explicit approval gates in agent and physical-arm VR interfaces.

Technical Skills

Languages and ML: Python, SQL, C++, PyTorch, XGBoost

Retrieval: BM25, dense retrieval, reciprocal rank fusion, BGE reranking, pgvector

Agents and Integration: LangGraph, MCP, WebXR, Python/UDP bridges

Systems and Evaluation: FastAPI, PostgreSQL, Docker, ONNX, paired-bootstrap intervals, GitHub Actions

Projects

MetaNavIT: Retrieval Evaluation and APIs | Python, PostgreSQL, pgvector, BM25, RRF, BGE

- Contributed retrieval/data APIs on a six-person team; extended the fork with hybrid retrieval, BGE reranking, and PostgreSQL/pgvector parity checks.
- Measured $+0.075$ paired nDCG@10 over BM25 on 300 SciFact queries (95% CI 0.053–0.101); generated evidence tables from committed results with drift checks.

Agentic Match Forecasting | XGBoost, LangGraph, MCP, FastAPI, Docker

- Connected XGBoost forecasts and contextual MCP tools through LangGraph/FastAPI, with calibration, conformal sets, and approval before staked recommendations.

DoM: Structurally Constrained Math Programs | Python, PyTorch, Transformers, MathQA

- Co-developed a decoder-only PyTorch transformer for MathQA computation graphs, including tokenization, training, and structurally constrained decoding.

SPUR: Metric Depth for Robotic Pruning | PyTorch, DINOv2, Blender, FastAPI, ONNX

- Recorded 0.0445 ± 0.0057 m depth RMSE with three stereo pairs on synthetic orchard data (five seeds; stored best-validation scores).
- Packaged Docker/FastAPI inference; measured the PyTorch refiner at 156 ms p50 (V100, fp16, six precomputed-depth views); checked separate ONNX export parity to $1.53e-5$.

Berkeley Humanoid Lite: Quest VR Teleoperation | Quest 2, WebXR, Python, inverse kinematics, SocketCAN

- Built Quest 2/WebXR teleoperation for simulated and physical Berkeley Humanoid Lite arms: Python/UDP pose bridge, inverse kinematics, and SocketCAN actuation.
- Integrated gravity feedforward, grip-based motion gating, and headset calibration; added camera, relative-depth/optical-flow, lidar, and IMU views.

Experience

Graduate Teaching Assistant, Oregon State University | Sep 2025 – Jun 2026

- Led 2-hour Operating Systems (C++) and Computational Methods (COMSOL) studios for 25+ students; taught debugging, concurrency, and numerical simulation.

Math and Computer Science Tutor, Oregon State University | Nov 2022 – Jun 2025

- Tutored 50+ students in mathematics and computer science; adapted debugging guidance and technical explanations to individual learning needs.

Education

Oregon State University, Corvallis, OR

Master of Science in Artificial Intelligence | Jun 2026

Bachelor of Science in Computer Science (Applied: Data Science); Mathematics Minor | Jun 2025